Calculus on Minkowski Spacetime

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Scope

These notes continue the geometry module and follow the third lecture in a more complete way. The focus is to connect:

- curves, fields, and observer kinematics,
- tensor/form language in Minkowski spacetime,
- integral and differential viewpoints,
- the geometric structure behind Maxwell theory.

The main message is that much of electrodynamics is geometry plus source content.

Why Use a Covariant View

Coordinate systems are useful, but they can hide structure. A covariant formulation keeps only geometric information that is physically meaningful.

In practice:

- coordinates are tools, not physics;
- observer choices change component values but not geometric objects;
- projection operators make observer dependence explicit and controlled.

This is especially important when moving between inertial observers or when comparing electric/magnetic splits.

Curves, Fields, and Their Relation

Keep the distinction clear:

- a **curve** is $x^\mu(\lambda)$,
- a **tangent** to that curve is $\mathrm{d}x^\mu/\mathrm{d}\lambda$,
- a **vector field** is $V^\mu(x)$ defined at all spacetime points.

Integral curves of a field are determined by

$$\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} = V^\mu(x(\lambda)).$$

So a field generates a congruence of curves, while a single curve gives one worldline.

> **Note**
>
> Using the same symbols in both contexts is common in physics, but conceptually they are not the same object.

Proper Time and Affine Parametrization

For timelike worldlines, proper time τ satisfies

$$\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = -1.$$

If $u^\mu \equiv \mathrm{d}x^\mu/\mathrm{d}\tau$, then

$$\eta_{\mu\nu}u^\mu u^\nu = -1.$$

This is the normalization associated with affine/proper-time parametrization. For non-affine parameters, this normalization is not constant in general.

In many calculations one can still parametrize by coordinate time, but proper time keeps covariance and normalization transparent.

Electromagnetic Split Relative to an Observer

Given Faraday tensor $F_{\mu\nu} = -F_{\nu\mu}$, define

$$E_\mu \equiv F_{\mu\nu}u^\nu, \qquad E_\mu u^\mu = 0,$$

and the spatial magnetic 2-form

$$B_{\alpha\beta} \equiv h_\alpha{}^\mu h_\beta{}^\nu F_{\mu\nu}.$$

Then

$$F_{\alpha\beta} = B_{\alpha\beta} + u_\alpha E_\beta - u_\beta E_\alpha.$$

So electric and magnetic fields are observer-dependent projections of one spacetime tensor, not two independent fundamental fields.

Differential Forms: Geometric Degrees

Useful interpretation by degree:

- 0-form: scalar field,
- 1-form: linear functional on vectors,
- 2-form: oriented area density,
- 3-form: oriented volume density,
- 4-form (in spacetime): oriented spacetime-volume density.

Wedge products encode oriented geometric content. Antisymmetry means repeated directions collapse to zero, which is exactly what should happen for oriented areas/volumes.

Exterior Derivative

For a scalar f,

$$\mathrm{d}f = (\partial_\mu f)\,\mathrm{d}x^\mu.$$

For a 1-form $\omega = \omega_\mu \mathrm{d}x^\mu$,

$$\mathrm{d}\omega = (\partial_\nu \omega_\mu)\,\mathrm{d}x^\nu \wedge \mathrm{d}x^\mu.$$

For a 2-form $\beta = \frac{1}{2}\beta_{\mu\nu}\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$,

$$\mathrm{d}\beta = \frac{1}{2}(\partial_\alpha \beta_{\mu\nu})\,\mathrm{d}x^\alpha \wedge \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

Key identity:

$$\mathrm{d}^2 = 0,$$

for smooth fields (commuting mixed partial derivatives).

Integrals of Forms and Parametrization Independence

A form is integrated on a geometric domain of matching dimension:

- 1-form on curves,
- 2-form on surfaces,
- 3-form on volumes.

The parametrization supplies tangent vectors and Jacobian factors. Under reparametrization, these compensate exactly, so the geometric integral is unchanged.

This is why one writes $\int_\Sigma \omega$ for a domain Σ itself, rather than tying meaning to one specific coordinate chart.

Stokes Theorem as the Unifying Principle

Prototype in 1D:

$$\int_a^b \mathrm{d}f = f(b) - f(a).$$

General form:

$$\int_{\partial\Omega} \omega = \int_\Omega \mathrm{d}\omega.$$

Classical theorems appear as special cases:

- gradient / fundamental theorem (1D),
- Green / Stokes (curve vs surface),
- divergence theorem (surface vs volume).

Thus, vector-calculus identities are different realizations of a single geometric statement.

Relation with standard integration

To make contact with usual Riemann (or Lebesgue) integration, we must understand how differential forms are integrated over domains.

Start with the 1D case. Let $I = [a, b] \subset \mathbb{R}$ and consider the 1-form:

$$\mathrm{d}f = \frac{\mathrm{d}f}{\mathrm{d}x}\mathrm{d}x.$$

The integral over I is defined via a parametrization of the interval. Introduce a parameter $\lambda \in [\lambda_0, \lambda_1]$ and a smooth map:

$$x = x(\lambda),$$

such that:

$$x(\lambda_0) = a, \quad x(\lambda_1) = b.$$

Pulling back the form to the parameter space can be understood as follows. A 1-form acts on tangent vectors, so along the parametrized curve $x(\lambda)$ we evaluate $\mathrm{d}f$ on the tangent vector:

$$\frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial}{\partial x}.$$

Thus,

$$\mathrm{d}f\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial}{\partial x}\right) = \frac{\mathrm{d}f}{\mathrm{d}x}(x(\lambda))\frac{\mathrm{d}x}{\mathrm{d}\lambda}.$$

Including the measure $\mathrm{d}\lambda$, the pullback becomes:

$$\mathrm{d}f \;\mapsto\; \frac{\mathrm{d}f}{\mathrm{d}x}(x(\lambda))\frac{\mathrm{d}x}{\mathrm{d}\lambda}\mathrm{d}\lambda.$$

Thus, the integral becomes:

$$\int_I \mathrm{d}f = \int_{\lambda_0}^{\lambda_1} \frac{\mathrm{d}f}{\mathrm{d}x}(x(\lambda))\frac{\mathrm{d}x}{\mathrm{d}\lambda}\mathrm{d}\lambda.$$

This is exactly the standard change-of-variables formula.

With the natural choice $x(\lambda) = \lambda$, we recover:

$$\int_a^b \frac{\mathrm{d}f}{\mathrm{d}x}\mathrm{d}x,$$

and therefore:

$$\int_a^b \mathrm{d}f = f(b) - f(a).$$

Higher-dimensional case

The same mechanism applies in higher dimensions, but now we must describe how a domain is parametrized and how forms act on tangent vectors.

Let Ω be a k-dimensional domain. Assume we have a one-to-one parametrization:

$$x = x(\lambda^1, \dots, \lambda^k),$$

with $(\lambda^1, \dots, \lambda^k)$ ranging over some region in $\mathbb{R}^k$, such that varying the parameters covers Ω.

If we vary one parameter at a time, we obtain k tangent vectors to the domain:

$$\frac{\partial x}{\partial \lambda^1}, \quad \dots, \quad \frac{\partial x}{\partial \lambda^k}.$$

These vectors span the tangent space of Ω at each point.

Now consider a k-form:

$$\omega = f(x)\,\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}.$$

To pull back ω, we evaluate it on these tangent vectors:

$$\omega\left(\frac{\partial x}{\partial \lambda^1}, \dots, \frac{\partial x}{\partial \lambda^k}\right).$$

Each 1-form $\mathrm{d}x^i$ acts on a tangent vector as:

$$\mathrm{d}x^i\left(\frac{\partial x}{\partial \lambda^b}\right) = \frac{\partial x^i}{\partial \lambda^b}.$$

Therefore, the wedge product produces:

$$\omega \;\mapsto\; f(x(\lambda))\det\left(\frac{\partial x^{i_a}}{\partial \lambda^b}\right)\mathrm{d}\lambda^1 \cdots \mathrm{d}\lambda^k.$$

The determinant arises from the antisymmetry of the wedge product: it encodes how the k tangent vectors combine into an oriented volume element. We will make this connection with the determinant more explicit below.

Finally, the integral becomes:

$$\int_\Omega \omega = \int f(x(\lambda)) \det\left(\frac{\partial x}{\partial \lambda}\right) \mathrm{d}^k\lambda,$$

which is precisely the usual Jacobian formula for multiple integrals.

Interpretation

- Differential forms encode both the integrand and the oriented measure.
- The pullback produces the Jacobian automatically.
- No additional structure is needed: change of variables is built into the definition.

Thus, Stokes theorem is not a new type of integration, but a geometric reformulation of standard integration in a coordinate-free language.

Worked Integration Examples

This section follows the lecture emphasis: concrete integral identities before moving to full Maxwell equations.

Example 1: Fundamental Theorem as 1D Stokes

Let $f(x) = x^3 - 2x$. Then

$$\mathrm{d}f = (3x^2 - 2)\,\mathrm{d}x.$$

On $[0, 2]$,

$$\int_0^2 \mathrm{d}f = \int_0^2 (3x^2 - 2)\,\mathrm{d}x = \left[x^3 - 2x\right]_0^2 = 4.$$

Boundary evaluation gives the same result:

$$f(2) - f(0) = 4 - 0 = 4.$$

Example 2: Green's Theorem from a 1-Form

In $\mathbb{R}^2$, take

$$\omega = P\,\mathrm{d}x + Q\,\mathrm{d}y \quad \text{with} \quad P = -y,\ Q = x.$$

Then

$$\mathrm{d}\omega = (\partial_x Q - \partial_y P)\,\mathrm{d}x \wedge \mathrm{d}y = (1 - (-1))\,\mathrm{d}x \wedge \mathrm{d}y = 2\,\mathrm{d}x \wedge \mathrm{d}y.$$

Let Ω be the unit disk. Stokes gives

$$\oint_{\partial\Omega} \omega = \iint_\Omega \mathrm{d}\omega.$$

Right side:

$$\iint_\Omega 2\,\mathrm{d}A = 2\pi.$$

Left side, with $x = \cos t$, $y = \sin t$:

$$\omega = -y\,\mathrm{d}x + x\,\mathrm{d}y = \sin^2 t\,\mathrm{d}t + \cos^2 t\,\mathrm{d}t = \mathrm{d}t,$$

so

$$\oint_{\partial\Omega} \omega = \int_0^{2\pi} \mathrm{d}t = 2\pi.$$

Example 3: 3D Stokes Theorem (Circulation-Curl)

Let

$$\mathbf{A} = (-y, x, 0),$$

with associated 1-form

$$\omega = -y\,\mathrm{d}x + x\,\mathrm{d}y.$$

Then

$$\mathrm{d}\omega = 2\,\mathrm{d}x \wedge \mathrm{d}y,$$

which corresponds to $\nabla \times \mathbf{A} = (0, 0, 2)$.

Choose Σ as the unit disk in the xy-plane, oriented upward. Then

$$\int_{\partial\Sigma} \omega = \int_{\Sigma} \mathrm{d}\omega = 2\pi.$$

This is exactly the circulation-curl theorem in form language.

Example 4: Divergence Theorem from a 2-Form

Let

$$\mathbf{B} = (x, y, z).$$

Associate the 2-form

$$\beta = B_x\,\mathrm{d}y \wedge \mathrm{d}z + B_y\,\mathrm{d}z \wedge \mathrm{d}x + B_z\,\mathrm{d}x \wedge \mathrm{d}y.$$

Then

$$\mathrm{d}\beta = (\nabla \cdot \mathbf{B})\,\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,$$

where we are using ∇ to represent the usual three-dimensional vector differential operator. Since $\nabla \cdot \mathbf{B} = 1 + 1 + 1 = 3$,

$$\iiint_V \mathrm{d}\beta = 3\,\mathrm{Vol}(V).$$

For the unit ball V, $\mathrm{Vol}(V) = \frac{4\pi}{3}$, so

$$\iiint_V \mathrm{d}\beta = 4\pi.$$

Hence, by Stokes in degree 2,

$$\iint_{\partial V} \beta = 4\pi,$$

which matches the classical outward flux of $\mathbf{B}$ through the unit sphere.

Example 5: Reparametrization Invariance (Curve Integral)

Take the 1-form

$$\omega = x\,\mathrm{d}x,$$

and the curve from $x = 0$ to $x = 1$.

Using parameter $t \in [0, 1]$, $x = t$:

$$\int_{\gamma} \omega = \int_0^1 t\,\mathrm{d}t = \frac{1}{2}.$$

Using parameter $s \in [0, 1]$, $x = s^3$:

$$\int_{\gamma} \omega = \int_0^1 s^3\,(3s^2)\,\mathrm{d}s = \int_0^1 3s^5\,\mathrm{d}s = \frac{1}{2}.$$

Same geometric curve, same oriented endpoints, same integral.

Divergence vs Curl in Form Language

In 3D Euclidean slices:

- from a 1-form, d gives a 2-form (curl-type content),
- from a 2-form, d gives a 3-form (divergence-type content).

The same operator d does both jobs, depending on degree. This is a major reason forms simplify electrodynamics.

Electric vs Magnetic Geometric Nature

Geometrically, in 3D:

- electric field is naturally a spatial 1-form,
- magnetic field is naturally a spatial 2-form.

The usual magnetic pseudovector is a 3D dual representation. It is convenient, but it hides that magnetism is fundamentally area-oriented.

Practical Takeaways

- Covariance helps separate geometric invariants from coordinate artifacts.
- Observer projection is the right way to discuss frame-dependent fields.
- Exterior derivative plus Stokes gives a unified differential/integral picture.
- The integral theorems are best seen as one geometric statement in different form degrees.

Volume Forms in Minkowski Spacetime

In Minkowski spacetime with Cartesian coordinates, the metric components $\eta_{\mu\nu}$ are constant. As a result, the covariant derivative reduces to the partial derivative, and the volume form can be characterized by the condition that its components are constant. First note that a volume form is a 4-form and can be written as

$$\mathbb{e}^4 = \frac{1}{4!}\mathbb{e}_{\mu\nu\alpha\beta}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu \wedge \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta = \mathbb{e}_{0123}\mathrm{d}x^0 \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3,$$

therefore we must have

$$\partial_\lambda\, \mathbb{e}^4_{\mu\nu\rho\sigma} = 0.$$

To fix the normalization, we require that in an oriented Cartesian basis,

$$\mathbb{e}^4(\partial_0, \partial_1, \partial_2, \partial_3) = \mathbb{e}^4_{0123} = +1.$$

This determines the spacetime volume form uniquely:

$$\mathbb{e}^4 = \mathrm{d}x^0 \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3.$$

Equivalently, this normalization corresponds to assigning unit volume to the coordinate hypercube $[0,1]^4$ (in the chosen units), so that its volume is one in those units (e.g., one meter to the fourth power).

More generally, this can be written in terms of the Levi-Civita symbol as

$$\mathbb{e}^4_{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma}.$$

Given a unit timelike vector u^μ, satisfying

$$\eta_{\mu\nu} u^\mu u^\nu = -1,$$

we can define a spatial volume form associated with the observer u^μ by contracting the spacetime volume form:

$$\mathbb{e}^3_u = i_u \mathbb{e}^4 = \mathbb{e}^4(u).$$

In components, this gives

$$\mathbb{e}^3_{u\alpha\beta\gamma} = u^\mu \, \epsilon_{\mu\alpha\beta\gamma}.$$

This object is a 3-form that is orthogonal to u^μ in the sense that

$$i_u \mathbb{e}^3_u = 0.$$

In the special case where $u^\mu = (1, 0, 0, 0)$, we obtain

$$\mathbb{e}^3_u = \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3,$$

which is the standard spatial volume form.

Thus, $\mathbb{e}^3_u$ represents the oriented volume element in the local rest frame of the observer u^μ. It provides the natural geometric object used to define spatial Hodge duals and to interpret quantities such as magnetic fields as area-like (bivector) objects.

The Hodge Star Operator (Geometric Definition)

The spacetime volume form $\mathbb{e}^4 \in \Lambda^4 T^*_p M$ provides a natural way to relate objects of complementary degree. In particular, it allows us to map vectors to 3-forms by contraction. Given a vector $v \in T_p M$, we define

$$i_v \mathbb{e}^4 = \frac{1}{3!} v^\mu \, \epsilon_{\mu\alpha\beta\gamma} \, \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta \wedge \mathrm{d}x^\gamma,$$

which is a 3-form with components $v^\mu \epsilon_{\mu\alpha\beta\gamma}$. This associates to each direction the oriented volume element orthogonal to it.

Using the metric, we can raise indices of $\mathbb{e}^4$ and define the corresponding contravariant object

$$\mathbb{e}_4 = \frac{1}{4!} \epsilon^{\mu\nu\rho\sigma} \, \partial_\mu \wedge \partial_\nu \wedge \partial_\rho \wedge \partial_\sigma,$$

which belongs to $\Lambda^4 T_p M$. This object acts on covectors and allows us to map 1-forms to 3-vectors:

$$\mathbb{e}_4(\tilde{w}) = \frac{1}{3!} w_\mu \, \epsilon^{\mu\nu\rho\sigma} \, \partial_\nu \wedge \partial_\rho \wedge \partial_\sigma.$$

More generally, the volume form defines natural maps

$$\Lambda^p T^*_p M \to \Lambda^{4-p} T_p M, \qquad \Lambda^p T_p M \to \Lambda^{4-p} T^*_p M,$$

by contraction. These maps exchange the degree of an object with its complementary degree.

Since the metric provides an identification between vectors and covectors, we can combine these constructions to obtain a map acting entirely within forms:

$$\star : \Lambda^p T^*_p M \to \Lambda^{4-p} T^*_p M.$$

This is the Hodge star operator. It is determined by the metric and the orientation, and encodes the duality between forms of complementary degree. More precisely, given a p-form $\alpha \in \Lambda^p T_p^* M$, we define

$$\star\alpha = i_{\alpha^\sharp}\, \mathbb{e}^4,$$

where $\alpha^\sharp$ is the p-vector obtained by raising the indices of α with the inverse metric η^{-1}. In components, this can be expressed (for a 1-form) as

$$\star\alpha = \frac{1}{3!}\alpha_\mu\, \epsilon^{\mu\nu\rho\sigma}\, \eta_{\nu\kappa}\eta_{\rho\lambda}\eta_{\sigma\tau}\, \mathrm{d}x^\kappa \wedge \mathrm{d}x^\lambda \wedge \mathrm{d}x^\tau.$$

For a 2-form $\beta \in \Lambda^2 T_p^* M$, the dual is again a 2-form,

$$\star\beta = \frac{1}{2!\,2!}\beta_{\mu\nu}\, \epsilon^{\mu\nu\rho\sigma}\, \eta_{\rho\kappa}\eta_{\sigma\lambda}\, \mathrm{d}x^\kappa \wedge \mathrm{d}x^\lambda.$$

In practice, for a 1-form $\tilde{w} = w_\mu \mathrm{d}x^\mu$, we obtain

$$\star\tilde{w} = \frac{1}{3!}w_\mu\, \epsilon^{\mu\nu\rho\sigma}\, \eta_{\nu\alpha}\eta_{\rho\beta}\eta_{\sigma\gamma}\, \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta \wedge \mathrm{d}x^\gamma = \frac{1}{3!}w^\nu\, \epsilon_{\nu\alpha\beta\gamma}\, \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta \wedge \mathrm{d}x^\gamma,$$

where $w^\nu = \eta^{\nu\mu}w_\mu$. This shows explicitly how the metric is used to convert between covariant and contravariant components in the construction of $\star$.

In an oriented orthonormal basis, $\star\alpha$ can be understood as the form obtained by completing the directions of α to a full oriented basis. This makes the action of $\star$ explicit in coordinates.

Explicit action in $\mathbb{R}^3$

In 3D Euclidean space with coordinates (x, y, z), the volume form is

$$\mathbb{e}^3 = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z.$$

The same construction used in four dimensions applies here. The Hodge star is determined by the requirement that it maps a form to the complementary directions needed to produce the volume form with the correct orientation.

For 1-forms:

$$\star\mathrm{d}x = \mathrm{d}y \wedge \mathrm{d}z, \quad \star\mathrm{d}y = \mathrm{d}z \wedge \mathrm{d}x, \quad \star\mathrm{d}z = \mathrm{d}x \wedge \mathrm{d}y,$$

For 2-forms:

$$\star(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}z, \quad \star(\mathrm{d}y \wedge \mathrm{d}z) = \mathrm{d}x, \quad \star(\mathrm{d}z \wedge \mathrm{d}x) = \mathrm{d}y.$$

For scalars and volume forms:

$$\star 1 = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z, \qquad \star(\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z) = 1.$$

These relations follow directly from the defining property that $\alpha \wedge \star\beta$ reproduces the volume form when α and β span complementary directions.

It is useful to relate this three-dimensional construction to the four-dimensional one. Given a unit timelike vector u^μ, a spatial vector v^μ (orthogonal to u^μ) can be represented as a bivector $u \wedge v$. Applying the four-dimensional Hodge star, we obtain

$$\star(u \wedge v) = \eta^{-1}(i_v i_u \mathbb{e}^4) = \frac{1}{2}u_\mu v_\nu\, \epsilon^{\mu\nu\alpha\beta}\, \partial_\alpha \wedge \partial_\beta,$$

which is a purely spatial bivector. This construction is coordinate-independent and projects out any component along u^μ.

In the frame where $u^\mu = (1, 0, 0, 0)$, this reduces to

$$\star(u \wedge v) = \frac{1}{2} v_i \, \epsilon^{0ijk} \, \partial_j \wedge \partial_k,$$

which reproduces the usual identification between spatial vectors and antisymmetric tensors in three dimensions. Thus, using $\mathbb{e}^4(u)$, we can define a Hodge dual acting on forms orthogonal to u:

$$\star_u : \Lambda^p_{\perp u} \to \Lambda^{3-p}_{\perp u}.$$

More explicitly, for a p-form $\tilde{w}$, we have

$$\star_u \tilde{w} \equiv \eta(\mathbb{e}_4(u, \tilde{w})).$$

Interpretation

- The 4D Hodge star is intrinsic to spacetime.
- The 3D Hodge star arises only after choosing an observer.
- This explains why operations like the cross product and the magnetic field depend on a choice of frame.

The Hodge star:

- uses the metric to identify orthogonal directions,
- uses orientation to fix signs,
- converts objects of dimension p into complementary objects of dimension $n - p$.

In 3D:

- 1-forms $\leftrightarrow$ 2-forms (vectors $\leftrightarrow$ pseudovectors),
- 0-forms $\leftrightarrow$ 3-forms (scalars $\leftrightarrow$ volume densities).

This is the mechanism behind:

- cross product,
- curl,
- divergence,

which arise from combining d and $\star$.

Differential Vector Operators

To introduce vector differential operators, we first relabel the basis covectors and vectors. We define

$$\gamma^\mu = \mathrm{d}x^\mu, \qquad \gamma_\mu = \partial_\mu,$$

so that $\{\gamma^\mu\}$ and $\{\gamma_\mu\}$ remain dual bases. This allows us to write the differential operator in compact form as

$$\partial \equiv \gamma^\mu \partial_\mu,$$

where ∂_μ is the covariant derivative (which reduces to the usual partial derivative in Cartesian coordinates), and γ^μ is treated as a constant basis 1-form.

With this notation, the antisymmetric (rotational) derivative of a vector field $v = v^\mu \gamma_\mu$ is

$$\partial \wedge v = \frac{1}{2} (\partial_\mu v_\nu - \partial_\nu v_\mu) \, \gamma^\mu \wedge \gamma^\nu,$$

while the divergence is

$$\partial \cdot v = \partial_\mu v^\nu \eta_{\mu\nu}.$$

Cross Product and Curl from Projection and Hodge Star

To recover the familiar three-dimensional operators, we introduce a reference unit timelike vector u and project onto the spatial subspace orthogonal to it. For two vectors v and w, the antisymmetric product projected along u is

$$u \wedge v \wedge w = u^\mu v^\nu w^\rho \, \gamma_\mu \wedge \gamma_\nu \wedge \gamma_\rho,$$

which represents a spatial oriented volume element. Applying the Hodge dual associated with u, we obtain a spatial vector:

$$\star_u(v \wedge w) = u^\mu v^\nu w^\rho \, \epsilon_{\mu\nu\rho\sigma} \, \gamma^\sigma.$$

In the frame where $u = (1, 0, 0, 0)$, this reduces to

$$\star_u(v \wedge w) = v^i w^j \epsilon_{0ijk} \, \gamma^k,$$

and using $\epsilon_{0ijk} = \epsilon_{ijk}$, we recover the usual cross product. Thus, the cross product is a special case of the wedge product combined with the Hodge dual. More precisely,

$$v \times w \equiv \star_u(v \wedge w).$$

Similarly, the curl can be written as

$$\partial \times v \equiv \star_u(\partial \wedge v) = u^\mu \epsilon_{\mu\nu\rho\sigma} \, \partial^\nu v^\rho \, \gamma^\sigma,$$

which again reduces, for $u = (1, 0, 0, 0)$, to

$$\partial \times v = \partial^i v^j \epsilon_{0ijk} \, \gamma^k = (\partial^2 v^3 - \partial^3 v^2)\gamma^1 - (\partial^1 v^3 - \partial^3 v^1)\gamma^2 + (\partial^1 v^2 - \partial^2 v^1)\gamma^3.$$

In this way, the standard three-dimensional vector operators arise from covariant four-dimensional constructions by projection along a chosen observer u.

Curl of a Bivector Field

Given a bivector field

$$B = \frac{1}{2} B^{\mu\nu} \gamma_\mu \wedge \gamma_\nu,$$

we define its curl by

$$\star_u \left(\partial \wedge B \right) = \frac{1}{2} u^\mu \, \epsilon_{\mu\nu\rho\sigma} \, \partial^\nu B^{\rho\sigma}.$$

This extends the notion of curl to bivector fields, which naturally appear in the covariant description of electromagnetism.

Since $\partial \wedge B$ is a spatial 3-form, the spatial Hodge dual produces a scalar:

$$\star_u \left(\partial \wedge B \right) = \frac{1}{2} u^\mu \, \epsilon_{\mu\nu\rho\sigma} \, \partial^\nu B^{\rho\sigma}.$$

In the frame where $u = (1, 0, 0, 0)$, this becomes

$$\star_u \left(\partial \wedge B \right) = \frac{1}{2} \partial^i B^{jk} \epsilon_{0ijk}.$$

Introducing the spatial dual vector

$$B^i = \frac{1}{2} \epsilon^{ijk} B_{jk} \; = \; (\star_u B)^i,$$

we obtain

$$\star_u \left(\partial \wedge B \right) = \partial_i B^i.$$

Thus,

$$\star_u (\partial \wedge B) = \partial \cdot (\star_u B).$$

That is, the curl of a bivector field is equivalent to the divergence of its Hodge dual vector field. This illustrates how the use of the Hodge star can hide the underlying geometric nature of the fields and operations. In three dimensions this identification is natural, but it arises from combining the wedge product with the Hodge dual and is not fundamental in four-dimensional spacetime. This is also a good example of how the Hodge star exchanges the exterior product and the divergence.

Codifferential Operator

The combination $\delta = \star \mathrm{d} \star$ is called the codifferential. It satisfies:

$$\delta : \Lambda^p T^* M \to \Lambda^{p-1} T^* M.$$

In coordinates, for a 1-form $\omega = \omega_i \mathrm{d}x^i$:

$$\delta\omega = \partial_\mu \omega_\nu \eta^{\mu\nu},$$

which is precisely the divergence.

Key property: $\delta^2 = 0$, analogous to $\mathrm{d}^2 = 0$.

Exterior Product as a Projection Operator

Another use of the exterior product is to define projection operators. For a vector v and a time-like vector u, one can compute exterior products like $v \wedge u$ to obtain a bivector. In the frame where $u = (1, 0, 0, 0)$, this bivector has components only in the spatial directions. Thus, the operation of wedging with u effectively projects out the temporal component of v and produces a spatial bivector. This is a powerful way to extract the spatial part of a vector field relative to a chosen observer.

To better see that we can compute $v \wedge u$ and then apply this to u, i.e.,

$$v \wedge u(u) = v \cdot u \, u - u \cdot u \, v = v + (v \cdot u)u = (\delta_\mu{}^\nu + u_\mu u^\nu) v^\mu \gamma_\nu = v^\mu h_\mu{}^\nu \gamma_\nu,$$

which is exactly the spatial projection we defined in geometry-review.qmd. It is also easy to show that

$$v = v \wedge u(u) - (v \cdot u)u,$$

that is, v can be written as the sum of its spatial projection and its temporal part. Therefore, the exterior product with u encodes the spatial projection operation in a geometric way. This is another example of how the wedge product can be used to define geometric operations without reference to coordinates.

Note that the same is true also for a 2-vector F. We know that $F \wedge u$ is a 3-vector that contains only components orthogonal to u. Applying this to u again gives

$$F \wedge u(u) = F(u) \wedge u + u \cdot u \, F = -F + F(u) \wedge u = -F^{\alpha\beta} h_\alpha{}^\mu h_\beta{}^\nu \gamma_\mu \wedge \gamma_\nu,$$

which is exactly minus the spatial projection of the bivector F onto the subspace orthogonal to u. Thus, just as for vectors, wedging with u and contracting with u again extracts the part of F that lives in the observer's rest space; the extra overall sign relative to the vector case simply reflects that the projector now acts on two indices instead of one. This is precisely the operation that isolates the purely spatial (magnetic) part of the electromagnetic bivector relative to the observer u, while the components with one leg along u encode the electric part.

Exercises

Exercise 1: The Electric/Magnetic Split Relative to an Observer (Essential)

a) Show directly from $E_\mu = F_{\mu\nu}u^\nu$ and antisymmetry of $F_{\mu\nu}$ that $E_\mu u^\mu = 0$.

b) Starting from the projector definition $B_{\alpha\beta} = h_\alpha{}^\mu h_\beta{}^\nu F_{\mu\nu}$, prove

$$F_{\alpha\beta} = B_{\alpha\beta} + u_\alpha E_\beta - u_\beta E_\alpha.$$

c) $B_{\alpha\beta}$ is antisymmetric and orthogonal to u^μ on both indices. Count its independent components in 4D spacetime and explain why this matches the 3 components of the familiar magnetic 3-vector.

Exercise 2: Closure and Worked Stokes Examples (Essential)

a) Compute $\mathrm{d}\omega$ for a generic 1-form $\omega = \omega_\mu \mathrm{d}x^\mu$ and verify explicitly, using commuting partial derivatives, that $\mathrm{d}^2\omega = 0$ in flat coordinates.

b) Repeat Example 2 (Green's theorem from a 1-form) using the square $[-1,1]\times[-1,1]$ instead of the unit disk, and confirm both sides still agree.

c) For $\mathbf{A} = (-y, x, 0)$ and a disk of radius R, compute $\int_{\partial\Sigma}\omega$ and $\int_\Sigma \mathrm{d}\omega$ explicitly and show both equal $2\pi R^2$.

d) For $\mathbf{B} = (x, y, z)$ and a ball of radius R, verify

$$\iint_{\partial V} \beta = 4\pi R^3.$$

Exercise 3: Volume Forms and Orientation

a) From $\mathbb{e}^4 = \frac{1}{4!}\mathbb{e}_{\mu\nu\rho\sigma}\mathrm{d}x^\mu\wedge\mathrm{d}x^\nu\wedge\mathrm{d}x^\rho\wedge\mathrm{d}x^\sigma$, show that antisymmetry implies only one independent component, $\mathbb{e}_{0123}$.

b) Show that exchanging two arguments of $\mathbb{e}^4$ flips its sign, and use this to compute ϵ_{0213}.

c) For a boosted observer $u^\mu = (\gamma, \gamma v, 0, 0)$, compute the components $\mathbb{e}^3_{u\,\alpha\beta\gamma} = u^\mu\epsilon_{\mu\alpha\beta\gamma}$ and confirm that $i_u i_u \mathbb{e}^4 = 0$ still holds.

Exercise 4: The Hodge Star in 4D — Checking the Signs (Essential)

a) Derive $\star(\mathrm{d}x^\mu)$ explicitly from the general 1-form formula $\star\tilde{w} = \frac{1}{3!}w^\nu\epsilon_{\nu\alpha\beta\gamma}\,\mathrm{d}x^\alpha\wedge\mathrm{d}x^\beta\wedge\mathrm{d}x^\gamma$.

b) Verify directly, by composing two Hodge stars, that $\star^2 = +1$ on 1-forms in signature $(-,+,+,+)$.

c) Verify $\star 1 = \mathbb{e}^4$ and $\star\mathbb{e}^4 = -1$.

d) Compute $\star(\mathrm{d}x^0\wedge\mathrm{d}x^1)$.

Exercise 5: The Hodge Star in 3D

a) Verify $\star\mathrm{d}x = \mathrm{d}y\wedge\mathrm{d}z$ using the relation between the Hodge star and ϵ_{ijk}.

b) Prove the identity

$$\epsilon^{ijk}\epsilon_{ilm} = \delta^j_l\delta^k_m - \delta^j_m\delta^k_l.$$

c) For $u = (1,0,0,0)$, compute $\star(u\wedge v)$ explicitly and confirm it reproduces the standard identification between a spatial vector and an antisymmetric tensor in three dimensions.

Exercise 6: Vector Differential Operators in the γ^μ Notation

a) Show that $\partial = \gamma^\mu \partial_\mu$ reproduces $\mathrm{d}f$ when acting on a scalar.

b) Compute $\partial \wedge v$ explicitly for $v^\mu = (ct, x, y, z)$, and explain why the result is zero (relate $v_\mu \mathrm{d}x^\mu$ to the exterior derivative of a scalar).

c) Show that $\partial \wedge \partial = 0$, assuming fields are smooth enough for derivatives to commute.

Exercise 7: Completing the Curl-of-a-Bivector Identity (Essential)

a) Starting from $\star_u(\partial \wedge B) = \frac{1}{2} u^\mu \epsilon_{\mu\nu\rho\sigma} \partial^\nu B^{\rho\sigma}$ and the spatial dual vector $B^i \equiv \frac{1}{2}\epsilon^{ijk} B_{jk}$, fill in the index manipulation the notes skip to derive $\star_u(\partial \wedge B) = \partial_i B^i$ explicitly.

b) Hence verify $\star_u(\partial \wedge B) = \partial \cdot (\star_u B)$.

c) Compute $\partial \wedge B$ for constant $B^{\mu\nu}$ and show that the curl of a constant bivector field vanishes.

Exercise 8: The Codifferential (Essential)

a) Using that d raises form degree by one and each $\star$ exchanges degree $p \leftrightarrow n - p$, show that $\delta = \star \mathrm{d} \star$ maps p-forms to $(p-1)$-forms.

b) Verify $\delta^2 = 0$ using $\mathrm{d}^2 = 0$.

c) Compute $\delta(\mathrm{d}f)$ for a scalar f and identify the result as the d'Alembertian $\Box f$.